5 TRANSACTIONS AND CONCURRENCY CONTROL

5.1 Transaction

A set of logically related operation to perform unit of work.

5.2 Degree of concurrency

The number of users (Transactions) using data bases simultaneously (concurrently).

5.2.1 Flat File System

In flat file system 1 file is a resource so, only one user allowed at a time.

5.2.2 DBMS File System

- In DBMS file system 1 record is a resource so, it allows as many users as the number of records.
- More degree of concurrency.

5.3 Main Operations in Transactions

5.3.1 Read (A)

Access the data item 'A' from DB file 'Disk' to programmed variable (main memory) in order to use current value of 'A' in transaction logic.

5.3.2 Write (A)

Modification of data item 'A' in DB file.

5.4 ACID Properties

To preserve integrity (correctness) each transaction must satisfy ACID properties

DBMS Software	A:Atomicity D:Durability } Recovery management component of DBMS software take cares of atomicity and durability	
	I : Isolation} Concurrency control component of DBMS software is responsible for isolation.	
DBA User	C : Consistency} User (DBA or DB developer) is responsible for consistency.	

5.5 Atomicity

Execute all operations of transaction including commit or execute none of the operation of transaction by the time of transaction termination.
Recovery management component should roll back, if transaction fails anywhere before commit.

5.5.1 Redo Operation

- It performs all the modification of database file because of transaction commit.
- Clean all the log entries of committed transaction.
- Redo is not performed after every commit but after every check point.

5.5.2 Undo Operation

- Undone all the write operation performed by the transaction.
- Convert dirty block (updated block) into clean block.

5.5.3 Check point

- Check point issued by DBMS software in regular interval.
- If checkpoint issued
  (I) Performs Redo operation on all the committed transaction until previous checkpoint.

Example :

9 : 00 AM	DB Started
:	
9 : 05 AM	1st checkpoint
:	
9 : 10 AM	2nd checkpoint
:	
9 : 15 AM	System Crash

5.5.4 System Crash

If System crash/failure happen, required operation to recover are
(I) All committed transaction until previous checkpoint will perform Redo.
(II) All uncommitted transaction in entire system will perform undo.
(III) Clean all log entries.